SHREERAM RS 241201051

Case Study: Evaluating Zoho Creator for Enterprise LCNC Development

System Foundation & Framework

Aim

Analyze enterprise LCNC platforms, focusing on Zoho Creator, to identify the optimal framework for rapid deployment.

Objective

- Evaluate LCNC models and enterprise development processes.
- Compare platform efficiencies, advantages, and limitations.
- Recommend a platform and deployment methodology.

Selected Application

Enterprise Resource & Asset Management System (ERAMS)

Overview

ERAMS manages operational assets, workflows, financial depreciation, requests, and analytics dashboards.

Functional Requirements

User & Role Management Module

- Registration, SSO, and Role-Based Access Control (RBAC).

Asset Tracking Module

- Enrollment, barcode/QR scanning, and location mapping.

Workflow & Service Request Module

- Ticket creation, automated routing, and status updates.

Financial & Analytics Module

- Depreciation calculation, budgeting, and executive reporting.

SHREERAM RS 241201051

Platform Evaluation & Selection

LCNC Platforms Analysis

LCNC Platform	Suitability	Advantages	Limitations
Bubble	Medium	Visual freedom for web MVPs	Limited backend scripting; weak native mobile support
AppSheet	Medium	Fast app creation from spreadsheets	Restricted UI; simple linear workflows
OutSystems	High	Full-stack enterprise architecture; native code	High licensing costs; requires developers
Microsoft Power Apps	High	Seamless M365 and Azure integration	Complex tiers; premium connectors needed for non-MS APIs
Zoho Creator	Very High	Hybrid LCNC; Deluge engine; auto-generates web/mobile	Proprietary Deluge language has initial learning curve

Why Zoho Creator is Best?

Reasons

- **Hybrid Balance:** Drag-and-drop tools combined with Deluge scripting for custom logic.
- **Cross-Platform Deployment:** Single build automatically generates native iOS, Android, and web apps.
- **Process Builder:** Blueprint enforces strict departmental approval workflows.

- **Ecosystem Integration:** Native connections to 1,000+ cloud applications.
- **Cost Efficiency:** Scalable enterprise security and governance at a lower TCO.

Development Process & Implementation Architecture

LCNC Application Development Process

Plaintext

```
None
Requirement Gathering
        ↓
Database Schema Design
        ↓
Visual Form & UI Building
        ↓
Deluge Logic & Workflow Setup
        ↓
API Integration & Webhooks
        ↓
Security & Role Assignment
        ↓
Testing & Quality Assurance
        ↓
Multi-Platform Deployment
        ↓
Governance & Continuous Improvement
```

Development Phases

1. Requirement Analysis

- Identify stakeholder needs, user personas, asset lifecycles, compliance guidelines, and security policies.
- **Deliverables:** SRS & Process Map.

2. System Design

- Design relational database schemas, UI wireframes, and approval state-machine maps visually.
- **Deliverables:** Database Schema & Role Permission Matrix.

SHREERAM RS 241201051

3. Development

- **Modules:** Asset Management, Approval Workflow Engine, Financial Calculation, Executive Dashboard.
- Tools: Drag-and-drop builders, Deluge script, HTML5/CSS3/JavaScript, relational cloud database.

Code snippet

```
None
// Automated Depreciation & Threshold Alert Script in Zoho Creator
asset_record = Asset_Form[ID == input.Asset_ID];
current_year = today.year();
purchase_year = asset_record.Purchase_Date.year();
age = current_year - purchase_year;
depreciation_rate = 0.15;
current_value = asset_record.Purchase_Price * (1 -
(depreciation_rate * age));

if (current_value < (asset_record.Purchase_Price * 0.20))
{
    asset_record.Status = "Replacement Recommended";
    sendmail
    [
        from: zoho.adminemail
        to: asset_record.Department_Head.email
        subject: "Asset Alert: " + asset_record.Asset_Name
        message: "Asset has reached minimum residual threshold
value."
    ];
}
asset_record.Calculated_Current_Value = current_value;
```

SHREERAM RS 241201051

Testing, Deployment & Maintenance Operations

4. Testing

Testing Types

- Functional form validation and Deluge path testing.
- Workflow trigger verification and RBAC security testing.
- User Acceptance Testing (UAT).

Example Test Cases

Test Case	Input / Condition	Expected Result	Verdict
Form Validation	Submit asset without Serial Number	Visual alert blocks submission	Pass
Deluge Script	Enter purchase date 5 years in past	Depreciation formula updates value	Pass
Workflow Approval	Manager clicks "Approve Maintenance"	Status updates; notification sent	Pass
Security Audit	Log in as Field Inspector	Sensitive fields are hidden/read-only	Pass
API Integration	Transmit payload via REST API	External ERP ingests JSON payload	Pass

5. Deployment

Deployment Options

- One-click web cloud and native iOS/Android distribution.

Cloud Infrastructure

- Multi-region cloud with automated SSL encryption and daily backups.

6. Maintenance

Activities

- Workflow optimization, user permission tracking, API monitoring, and feature enhancement.

Comparative Metrics, Results & Strategic Conclusion

Comparison of LCNC Platforms

Criteria	**Bubble**	**AppSheet**	**OutSystems**	**Power Apps**	**Zoho Creator**
Requirement Changes	Medium	Poor	Excellent	Excellent	**Excellent**
Feedback Speed	Very High	Very High	Medium	High	**Very High**
Risk & Governance	Medium	Low	High	High	**High**
Delivery Speed	Fast	Very Fast	Medium	Fast	**Very Fast**
TCO	Low	Low	High	Medium	**Competitive**
Suitable for ERAMS	No	Partially	Yes	Yes	**Best**

SHREERAM RS 241201051

Results

- ERAMS needs continuous workflow adaptation, custom logic, and cross-platform availability.
- Traditional coding introduces extended timelines and backlogs.
- Pure no-code platforms lack support for complex relational schemas.
- Heavy low-code options carry high costs and demand specialized full-stack skills.
- Zoho Creator delivers the ideal hybrid balance, making it most suitable for enterprise apps.

Conclusion

Selecting a platform depends on organizational needs. For dynamic enterprise systems like ERAMS, Zoho Creator provides flexibility, cross-platform deployment, automated governance, and accessible low-code scripting. This combination ensures faster feature delivery, lower overhead, and improved business agility.